THE MALAWI NATIONAL EXAMINATIONS BOARD

2022 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

CHIEF EXAMINER'S REPORT

BIOLOGY (M022)

PAPER I

The 2022 Biology paper I was well set with questions ranging from low to high order which measured different skills. The paper covered the syllabus well. All the diagrams were clear enough. There were no typographic errors. The general performance was average.

A. GENERAL COMMENTS ON:

Knowledge of Subject Matter

Candidates rated poorly on knowledge of subject matter as they failed to: draw a genetic cross in **question 3 c,** take hydrophytes as fish in **question 5 b (ii),** failed to draw a table in **question 7,** take Pavlov as an animal in **question 11,** describe blood transfusion instead of blood clotting in **question 12.**

Reasoning Ability

Candidates were also rated poorly in reasoning ability as they: failed to apply knowledge of theory of evolution **in question 4 a,**

Candidate's Proper Use of Vocabulary

Candidates failed to write some words in English e.g. "sifilisi" instead of syphilis in **question 6 b (i),** "belu" instead of bell in **question 11**.

- Misspelt some words e.g. drug instead of drag in question 3b
- Used fart instead of fat
- Used God instead of dog in question 11

Ability to Make Accurate Observations

The candidates were rated poorly as they failed to observe **Figure 1** on **question 2**. Taking it as part of reproductive system instead of spinal cord; Failed to observe five organisms to make a food chain form the food web in question **3 a (i);** Failed to observed **Figure 3** on **question 8**. Taking **part Y** as a storage organ

Ability to Define Terms

The candidates failed to define residual air in **question 1 a (i).**

Application of Basic Principles and Make Relevant Conclusions

This was rated poorly as candidates failed to use the given letters to represent alleles when constructing a genetic cross in **question 3c.**

B. COMMENTS ON GENERAL PERFORMANCE OF CANDIDATES ON THE WHOLE PAPER

In general, the performance of candidate in 2022, Biology I, was poor. They failed to manipulate figures e.g. **question 7a** where they failed to calculate the mean; failed to make drawings e.g. **question 3 a (i)** candidates failed to draw arrow in food chain and draw lines to show fertilisation; inability to interpret photographs e.g. **question 2 (i)** candidates took the spinal cord as part of the reproductive system.

D. COMMENTS ON PERFORMANCE OF CANDIDATES ON INDIVIDUAL QUESTIONS

Question 1 a (i)

The question asked candidates to define residual air. The performance was poor. The expected response was air that remains in the lungs after a heavy exhalation. However, some candidates defined it as air that is breathed out.

Question 1 a (ii)

The question asked the candidates to state the importance of residual air in the lungs. It was poorly performed. The expected answer was to maintain shape of the lungs but most of the candidates but most candidates left it an attempted.

Question 1 b

The question asked the candidate to explain any one way through which the skin regulates body temperature. This was poorly done. One of the expected responses was contraction of the hair erector muscle raises the hairs to help trap a layer of still air. Candidates mentioned cilia instead of hairs. Others mentioned sores instead of goose pimples.

Question 2 a (i)

The candidates were asked to name the parts labelled D and F in a figure 1. At least a candidates were able to mention one of the two. The expected response was: D was grey matter and F was ganglion. However, some candidates named D as stomata while F as guard cell.

Question 2 a (ii)

The candidates were asked to mention any two functions of the fluid found in part labelled E in Figure 1. This was poorly performed. The expected responses were nourishing cells of the spinal cord and removing metabolic wastes from the spinal cells. The wrong answers from some candidate included lubricating the vagina among others.

Question 2 b

The question required the candidates to give any two functions of the spinal cord. This was poorly performed. The expected responses were linking the peripheral nervous system to the brain and coordinates reflex actions in the body. Most candidates were confusing functions of spinal cord and the spine.

Question 3 a (i)

The question asked the candidates to construct a food chain with five organisms from a given food web, in a figure. This was the most popular question but most candidates got a mark out of the allocated 5 marks. The expected answer was

green plants ⟶ grasshoppers ⟶ lizards ⟶ snake ⟶ hawk. However, some candidates did not include the arrows instead used mere lines.

Question 3 a (ii)

The question asked the candidates to name the trophic level occupied by toads in a diagram of a food web. The expected response was third trophic level although most candidates named it as secondary consumer.

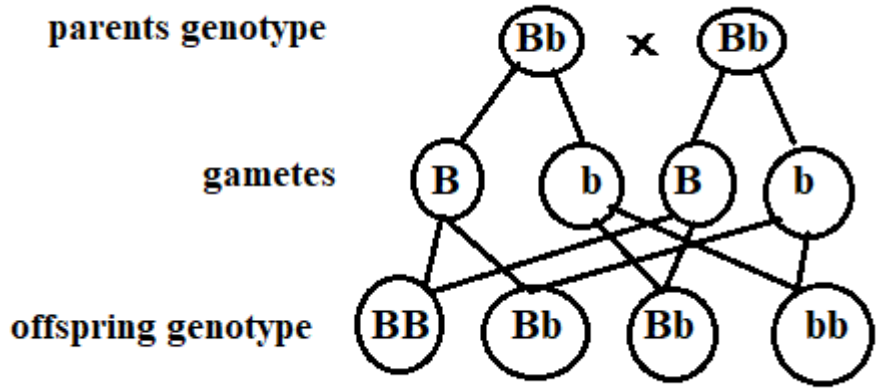
Question 3 b

The question asked candidates to describe any one adaptation of fish to locomotion. This was poorly done. Among the expected response was fish have streamlined shape to reduce drag. Most candidates failed to explain and also used wrong spellings, for example drug instead of drag.

Question 3 c

The question asked candidates to make a genetic cross to show the results after mating the F_1 generation**.**

The expected response was as below:



However, most candidates constructed a cross involving pure breeds while some were using the own letters instead of the given B and b.

Question 4 a (i)

The question asked candidates to explain what was causing the differences in populations between the two types of snakes (green and black mambas). It was poorly done by most candidates. They considered the green mambas feeding on the hawks and multiply while others indicated that the snakes fought each other, then the defeated ones were eaten by the hawks. Some even said the green mamba carried out photosynthesis and multiply. The expected response was the black mambas were easily seen by the hawks which preyed on them thereby reducing their numbers.

Question 4 a (ii)

The question asked the candidates to state the term that is used to describe the situation in **4 a (i)** above. This was fairly done. The correct answer was natural selection. Those who failed did not attempt the question.

Question 4 a (iii)

The question asked candidates to mention any two causes of speciation. It was fairly done. The expected response were natural selection and isolation (geographical, ecological, behavioural, reproductive). However, some candidates wrote natural speciation instead of natural selection

Question 4 b

The question asked candidates to state any one application of biotechnology in agriculture. This was poorly done. Among the expected answers was production of new varieties of crop. Candidates wrote application of manure or farm machinery which were off points.

Question 5 a

The question asked the candidates to explain any one way in which glucose is used after photosynthesis. This was fairly done. They were able to mention; for respiration, it is stored etc. However most failed to give the supporting point to the stated answer. While others responded the question in relation to use of glucose after digestion.

Question 5 b (i)

The question asked the candidate to describe any one type of water plants. This was poorly done. Among the expected answers was; submerged plants which completely grow in water. Candidates described specific examples of water plants such as water hyacinth, reeds etc. instead of types**.**

Question 5 b (ii)

The question asked candidates to explain any one adaptation of hydrophytes to their habitat. This was poorly done. One of the expected answers was that they have air spaces to maintain buoyancy. Candidates instead, regarded hydrophytes as fish hence gave answers as having streamlined shape, have scales etc. which were off point.

Question 6 a (i)

The question asked candidates to state types of immunity. It was fairly done. The expected answers were natural immunity and artificial immunity. However, some candidates were giving their answers as physical, exotic, animal, sexual and asexual which were not correct.

Question 6 a (ii)

The candidates were asked to name the type of immunity a baby gets from breast feeding. It was poorly done. The expected answer was natural passive immunity. Some candidates did not specify as passive while some gave answers such as Lilongwe dairy.

Question 6 a (iii)

The candidates were asked to explain the characteristic of the immunity mentioned in **6 a (ii).** It was poorly done. The expected answer was it is short live because antibodies are not produced in the body of the baby. Some candidates wrote it has colostrum.

Question 6 b (i)

The candidates were asked to mention any one sexually transmitted infection caused by bacteria. This was fairly done but some candidates used Chichewa such as “sifilisi” and wrong spellings as “ghonoria”

Question 6 b (ii)

The candidates were asked to state any one effect of sexually transmitted infections on an individual. It was fairy done. The expected answers include death, low productivity, infertility etc. but some candidates mentioned school dropout.

Question 7 a

The candidates were asked to present the given information in the question stem in a table form. The expected answer was as tabulated below

Mass of babies (kg)	Number of babies
1.5	5
1.8	6
2.5	3

It was poorly done as most candidate interchanged the independent and dependent variables. Some drew histograms instead of table.

Question 7 b

The candidates were asked to calculate the mean mass of the babies. It was poorly done. The expected answer was $\frac{(1.5\times5)+(1.8\times6)+(2.5\times3)}{14} = \frac{25.4}{14}$ = 1.8kg. Most candidates calculated the median. Some even simply added 1.5, 1.8 and 2.5 then divided by 14.

Question 8 a

The candidates were asked to identify the micro-organism in a diagram. The expected answer was paramecium. It was poorly done. Most off points answers included chishango, penicillium and protozoa.

Question 8 b

The question asked the candidates to name the part labelled X in a given diagram. The expected response was food vacuole. It was poorly done. Some off point responses were star and eye.

Question 8 c

The question asked the candidates to name the habitat of the organism in **Figure 3**. The expected response was aquatic or water. It was fairly done Some off point responses were terrestrial habitat and damp habitat

Question 8 d

The question asked the candidates to mention the function of the part labelled Y in figure 3. The expected response was controls all activities in the organism. It was fairly done. Some off point responses were storage of food and control the organism.

Question 9

The question asked the candidates to explain two problems that may arise if maternal and foetal blood mixes during pregnancy. This was poorly done. Expected responses were; it may lead to blood agglutination if the blood groups are incompatible and high blood pressure in the maternal blood may lead to bursting of foetal blood vessels. Some candidate wrote the child will suffocate, the child will drink blood etc.

Question 10 a (i)

The question asked the candidates to explain one way in which comparative anatomy provides evidence of evolution**.** It was poorly done. The expected response was that comparing structures of different vertebrates show some similarity suggesting their common ancestry. Some candidates defined an atom while others used prehistorical organisms and compared with current organisms.

Question 10 a (ii)

The question asked the candidates to explain one way in which embryology provides evidence of evolution. It was poorly done. The expected response was that comparing the embryos of different vertebrates shows some similarities which suggest their common ancestry. Most candidate were simply defining embryology.

Question 10 b

Candidates were asked to describe one evidence that show natural selection in action. It was fairly done. Among the expected responses is sickle cell anaemia where individual who are homozygous recessive to it do no rich maturity. Some candidates were copying information relating green and black mambas in question **4** (**a**).

Question 11

Candidates were asked to describe how Pavlov showed conditioned reflex using his experiment on dogs. The expected response was an orderly presentation of steps that Pavlov did to condition the dogs to respond to ring of the bell place of food. It was poorly done as most candidates considered Pavlov as an animal that can be called. Some used a whistle instead of a bell that was used by Pavlov.

Question 12

The question asked the candidates to describe the process of blood clotting. It was poorly done. The response required the orderly presentation of events and substances involved in the blood clotting process. Most candidates failed to follow the order while others confused it with blood transfusion.

Question 13

Candidates were asked to describe any five ways of controlling cancer. It was fairly done. Most candidate failed to support their stated points. Some of the expected responses were reduce intake of fat rich food to reduce fat deposit in the body, do not smoke to prevent tar which triggers cancer, do physical exercises to burn the fats, eating food rich in vitamin E and C which reduce abnormal multiplication of cells, and work to reduce stress. Some funny answers were avoid sleeping with men who have big penis, avoid drinking and swallowing sperms and cutting the penis.

PAPER II

A. GENERAL COMMENTS ON THE QUALITY OF THE PAPER

The 2022 practical paper was generally good in the following aspects: The paper assessed relevant skills required by a practical paper like drawing skills, observation skills, manipulation of apparatus and deductions from observations among others. The questions accommodated all the levels of Bloom's Taxonomy from low to high order. The language used was appropriate for the

level of MSCE candidates. Candidates were given adequate spaces for responses however a few who made errors opted to use the blank pages at the back which should be discouraged. The questions were clear and gave proper instruction for the candidates to give the required responses. The paper was also good as it had no carry over questions.

B. GENERAL COMMENTS ON OVERALL PERFORMANCE OF CANDIDATES

The general response of the candidates was below average which is lower than the previous year. The majority scored below 15 out of the total 40 marks. This has been attributed to poor practical skills attained by the candidates in secondary education.

C. PERFORMANCE OF CANDIDATES ON INDIVIDUAL QUESTIONS

SECTION A

ESSAY QUESTIONS

This section contains two question each worth 10 marks. The essays involve describing an experiment that candidates can conduct in various topics to show specific biological facts. These essays were not supposed to have heading, numbers or bullets but the candidates should describe what should be done in short paragraphs.

Question 1

Candidates were required to describe an experiment that can be conducted to show that germinating bean seeds contain an enzyme that can digest starch. In their responses candidates should have included the following: First crush the bean seeds and add water to obtain the enzyme/extract. Put starch solution in test tubes labelled A and B. Then add enzyme extract to the tube A only. Leave them for a few minutes to allow

enzyme to work. Then add iodine to both test tubes to test for starch. Observe colour change. It is expected that tube A will show brown colour and B show blue-black colour. This will prove that germinating bean seeds contained enzymes that digested starch.

However, candidates performed poorly in this question as most candidates failed to show how to extract the enzyme. A good number of candidates also failed to include starch as a substrate and they were only testing the presence of starch in the bean seeds themselves. Others were unaware that plants also contain enzymes and they could include salivary amylase as the enzyme.

Question 2

Candidates were required to describe an experiment that can show that physical exercise affects breathing rate. The question was attempted by the majority of the candidates and the performance was fairly good as most candidates were able to link physical activity and breathing rate.

The correct response was; Measure the breathing rate or count the number of breaths in one minute of the person who is at rest. Record the result. Allow the same person to do a physical exercise for a while. Count the number of breaths in one minute soon after the exercise. Then compare the two values and after physical exercise the one will show a high value while one before exercise will be lower.

The results will show that physical exercise increases breathing rate. Despite, most candidates who attempted the question were unable to differentiate breathing rate from pulse rate or heart beat rate, and they used the terms interchangeably. Some brought in effects of physical exercise on breathing like oxygen concentration, CO_2 concentration, breathing depth etc. Others measured wrong variables like distance, time temperature instead of counting number of breaths.

SECTION B

This section contained questions which required use of specimens provided to get correct responses. It was noted that despite candidates having the specimens they provided responses from what they learned even though the specimens provided lacked that information or aspects.

Question 3 a

The question was accompanied by specimen **M** which was groundnut flour, **N** which was maize flour, 50 ml iodine solution, 1 solution dropper. Candidates were asked to conduct an experiment to show the presence of starch in the specimens in a table provided. Many candidates attempted the question and they performed well. Most candidates were able to add iodine to the specimens and obtain the correct results as required as follows:

Procedure	**Observation**	**Conclusion**
Put specimen on the paper or any container provided Add iodine solution Observation colour change	Colour changed to blue-black	**M** Contains starch

This was for both specimens. However, most candidates were unable to state the required procedural steps using the materials provided as they started by crushing the specimen which was already in powder form, this led to writing unnecessary steps which resulted in shortage of space in the table. Other candidates did not know that iodine is used to test for starch which showed that students do not perform experiments during lessons. Some candidates were unable to differentiate observation and conclusion. A good number also gave the result for groundnuts as brown after

adding iodine proving that they just remembered what was taught in class while in experiment ground nut flour contains some starch.

Question 3 b

The question asked the candidates to give a reason for using the samples in powder form. This was very poorly performed. Very few candidates got it correct. The response was "to increase surface area for fast reaction of iodine".

However, many candidates failed to reason in that way and gave wrong responses, like for easy observation of colour change; iodine dissolves faster etc.

Question 4 a

Specimens provided were **P** – (down feather) **Q** – (flight feather). Candidates were asked to identify the specimens. The performance was very good for this question as most candidates were able to give correct names with a few who gave wrong spelling like 'dawn' for down, 'flying' for flight and upstroke and down stroke for flight and down feathers respectively.

It was regrettable to note that in some centres the subject teachers mismatched specimen with labels which made the candidates to respond wrongly.

Question 4 b

The question asked candidates to draw flight feather and label any three parts. The performance was very poor because instead of drawing a feather most candidates drew a leaf by the look of the apex and veins instead of barbs. The correct drawing should have had a sharpened quill, round "apex" and barbs not veins.

Question 4 c

The question asked for observable differences between specimen **P** and **Q**. The performance of candidates was fair; most candidates gave the correct responses which included. **P** has no after shaft while **Q** has after shaft, **P** has a weak shaft while **Q** has a strong shaft. **P** has loose barbs while **Q** has interlocked barbs.

However, a few failed to match the different aspects like **P** has a small shaft while **Q** has a strong shaft. Others just gave one side of the response like **P** has no after shaft. Some gave functional differences which were not observable.

Question 4 d

The question asked for an adaptation of flight feather **(Q)** for locomotion. Candidate's performance was fair. The correct response(s) included "it is hollow to reduce weight" or "it is broad to increase surface area for lift." Many candidates managed to state the feature but a lot failed to explain how it aids in locomotion. Some gave wrong spellings like harrow, horror, horror instead of hollow. Others failed to relate a feature and its use e.g. its broad to reduce weight.

RECOMMENDATIONS

(i) Schools Should

- cover the syllabus adequately and thoroughly.
- provide adequate teaching and learning resources to teachers and students e.g. charts, books etc.
- give correct specimens to candidates with correct labelling to avoid misleading candidates to wrong responses (Subject teachers).
- not leave placement of specimen to caretakers
- conduct INSETs to improve teaching and learning of science subjects.
- give students opportunities to conduct experiments where required (Subsect teachers)

- Make Chief Examiners reports accessible to teachers and encourage them to put into use the experiments to improve results. (Head teachers)
- assess students regularly (Subject teachers).
- teach Biology in English (Subject teachers)
- advise students to use legible handwriting (Teachers)

(i) Ministry of Education should

- make deliberate and sustainable efforts to supply laboratory equipment and materials to mostly Community Day Secondary Schools in rural areas
- deploy well qualified Secondary School Science Teachers to mostly Community Day Secondary Schools in rural areas to teach science subjects
- should provide enough financial resources through monthly ORT and Quarterly grants to schools to purchase enough teaching and learning materials including laboratory chemicals.